

Theory Week 9

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Mechanics 3 CAB G59
WhatsApp group



- Find all docs on my website (link in WhatsApp bio)
- remember to do the STACK
- Study Center
→ Wednesday 18:00-20:00 (ETC 4)

9	11/11/24	moving frames and inertial forces	3.4.2-3.4.3
	13/11/24	inertial forces (continued), moving-frame examples	3.4.3
exercise 9: rotating frames, apparent forces			

Balance of lin. momentum

Assume: $\square$ frame C has a fixed cartesian basis $\{\underline{e}_1^C, \underline{e}_2^C, \underline{e}_3^C\}$ & origin O^C

$\square$ frame M has a rotating basis $\{\underline{e}_1^M, \underline{e}_2^M, \underline{e}_3^M\}$ & moving origin O^M

time derivative relation between an inertial frame C and a moving frame M :

rotating: $\dot{\underline{r}}^C = \dot{\underline{r}}^M + \underline{\Omega}^M \times \underline{r}$, general: $\dot{\underline{r}}_P^C = \dot{\underline{r}}_{O^M}^M + \dot{\underline{r}}_{O^M P}^M + \underline{\Omega}^M \times \underline{r}_{O^M P}$

From position relation:

take $\frac{d}{dt}$ & do math

$$\ddot{\underline{r}}_P^C = \ddot{\underline{r}}_{O^M}^M + \underbrace{\frac{d\underline{\Omega}^M}{dt} \times \underline{r}_{O^M P}^M}_{\text{Euler acceleration}} + \underbrace{2\underline{\Omega}^M \times \dot{\underline{r}}_{O^M P}^M}_{\text{Coriolis acceleration}} + \underbrace{\underline{\Omega}^M \times (\underline{\Omega}^M \times \underline{r}_{O^M P}^M)}_{\text{centripetal acceleration}}$$

absolute acc.
 as seen from outside - absolute
 rel. time derivative (as seen from rotating frame)
 ang. vel. of M-coordinate sys.

$$(\underline{r}_{O^M P}^M = \underline{r}_{O^M}^M + \underline{r}_{O^M P}^M)$$

$\Rightarrow$ LMB in rot. frame:

balance of linear momentum in a moving frame M for a particle:

$$m\underline{a}^M = \underline{F}_{\text{ext}} + \underline{F}_{\text{Coriolis}} + \underline{F}_{\text{Euler}} + \underline{F}_{\text{centrifugal}} - m\underline{a}_{O^M}^M$$

fictitious force due to moving origin

Coriolis, Euler and centrifugal forces:

$$\underline{F}_{\text{Coriolis}} = -2m\underline{\Omega}^M \times \underline{v}^M, \quad \rightarrow \text{only when moving relative to frame } (\underline{v}^M \neq 0)$$

$$\underline{F}_{\text{Euler}} = -m \frac{d\underline{\Omega}^M}{dt} \times \underline{r}^M, \quad \rightarrow \text{only when } \underline{\Omega}^M \neq \text{const}$$

$$\underline{F}_{\text{centrifugal}} = -m\underline{\Omega}^M \times (\underline{\Omega}^M \times \underline{r}^M) \rightarrow \text{in 2D: } M(\underline{\Omega}^M)^2 \underline{r}^M, \text{ always acts in rot. sys.}$$

remember right hand rule for cross-products! (also the signs of cross-product)